

Bassel Saleh

bassel@utexas.edu

1000 San Marcos St, Apt 168, Austin TX 78702
(281) 515-2772

Education

Doctor of Philosophy December 2024 (Expected)
Computational Science, Engineering, and Mathematics (CSEM)
The University of Texas at Austin, Oden Institute

Master of Science December 2020
Computational Science, Engineering, and Mathematics (CSEM)
The University of Texas at Austin, Oden Institute

Bachelor of Science, Physics May 2018
The University of Texas at Austin, College of Natural Sciences

Bachelor of Science, Computer Science May 2018
The University of Texas at Austin, College of Natural Sciences

Research Experience

Graduate Research Assistant, The University of Texas at Austin August 2018 - Present
Oden Institute for Computational Engineering and Sciences
OPTIMUS Group
Advisor: Omar Ghattas, Ph.D.

Undergraduate Research Assistant, Cornell University June 2017 – August 2017
Laboratory for Accelerator-based Sciences and Education (CLASSE)
NSF Research Experience for Undergraduates
Advisor: Gennady Shvets, Ph.D.

Undergraduate Research Assistant, The University of Texas at Austin June 2016 – May 2018
Oden Institute for Computational Engineering and Sciences
Moncrief Summer Internship, Extended
Advisor: Omar Ghattas, Ph.D.

Undergraduate Research Assistant, The University of Texas at Austin. January 2015 – April 2016
Department of Computer Science
Advisor: Etienne Vouga, Ph.D.

Teaching Experience

Graduate Teaching Assistant, The University of Texas at Austin Fall 2024
Graduate course: Introduction to Mathematical Modeling in Science and Engineering I

Mentor, STEM MUSE Program Fall 2023 – Spring 2024
Provided academic and career guidance for two undergraduates from underrepresented gender identities

Mentor, SIAM Applied Math Mentorship Program Fall 2021 – Spring 2022
Provided guidance and taught advanced concepts to an undergraduate in mathematics

Instructor, Activateen March 2019 – May 2019
Led a workshop on Machine Learning Basics for Teens

Instructor, Activateen July 2018 – July 2018
Led a workshop on Introduction to Python and Programming Basics

Instructor, GT Learning Academy July 2015 – August 2015
Taught summer courses on algebra II, precalculus, chemistry, and physics for high schoolers

Publications and Technical Reports

Preprint (in review)

B. Saleh, A. Zimmerman, P. Chen, O. Ghattas. “Tempered Multifidelity Importance Sampling for Gravitational Wave Parameter Estimation.” (2024) [arXiv](#)

Preprint

B. Saleh, A. Zimmerman, D. Luo, O. Ghattas. “Improving Importance Sampling Efficiency with Bayesian Approximation Error.” (2024)

Preprint

B. Saleh, T. O’Leary-Roseberry, B. Keith, O. Ghattas. “Noise Aware Neural Operators for Bayesian Inverse Problems.” (2024)

Technical Report for Turing Honors Thesis

B. Saleh, O. Ghattas. “Scientific Machine Learning: A Neural Network-based Estimator for Forward Uncertainty Quantification.” UTCS Technical Reports. (2018) [Link to report](#)

Final Report for Cornell CLASSE REU

B. Saleh, G. Shvets, V. Khudik, T. Wang. “Modeling Bubble Formation in Plasma-Based Particle Accelerators.” CLASSE REU Reports. (2017) [Link to report](#)

Presentations and Posters

Oral Presentation

April 2024

American Physical Society April Meeting

Session on Gravitational Wave Parameter Estimation I: Methods

B. Saleh, A. Zimmerman, P. Chen, O. Ghattas. “Tempered Multifidelity Importance Sampling for Gravitational Wave Parameter Estimation.”

Poster

April 2023

American Physical Society April Meeting

B. Saleh, T. O’Leary-Roseberry, O. Ghattas, B. Keith. “Goal-Oriented Neural Network Surrogates for Gravitational Wave Models”

Oral Presentation

April 2022

SIAM Conference on Uncertainty Quantification

*Mini-symposium on Physics-Informed and Data-Driven Predictive Models with Quantified Uncertainty*B. Saleh, T. O’Leary-Roseberry, B. Keith, O. Ghattas. “Parametric Machine Learning Surrogates for Gravitational Wave Signals.”**Poster**

November 2021

IPAM Workshop on Source Inference and Parameter Estimation in Gravitational Wave Astronomy

B. Saleh, A. Zimmerman, P. Chen, O. Ghattas. “Multifidelity Importance Sampling for Gravitational Wave Inference.”**Oral Presentation**

March 2021

SIAM Conference on Computational Science and Engineering

*Mini-symposium on Computational Strategies for High Dimensional Stochastic Problems*B. Saleh, A. Zimmerman, P. Chen, O. Ghattas. “Multifidelity Importance Sampling for Gravitational Wave Inference.”**Oral Presentation**

April 2020

American Physical Society April Meeting

*Session on Gravitational Wave Analysis*A. Leviyev, B. Saleh, J. Chen, P. Chen, O. Ghattas, A. Zimmerman. “Stein Variational Inference for Gravitational Wave Likelihood Estimation.”**Oral Presentation**

March 2020

SIAM Conference on Uncertainty Quantification*

*Mini-symposium on Reduced Order Methods for Uncertainty Quantification in CFD Parametric Problems*B. Saleh, T. O’Leary-Roseberry, O. Ghattas. “Neural Networks as Control Variates for UQ in Ice Sheet Flow.”**Cancelled due to Covid-19 pandemic***Poster**

March 2017

SIAM Conference on Computational Science and Engineering

B. Saleh, U. Villa, O. Ghattas. “Neural Networks as Reduced Models for Physical Systems and Inverse Problems.”**Oral Presentation**

August 2017

CLASSE REU

B. Saleh, G. Shvets, V. Khudik, T. Wang. “Modeling Bubble Formation in Plasma-based Particle Accelerators.”**Poster**

August 2016

UT Summer Research Scholars Poster Session

B. Saleh, U. Villa, O. Ghattas. “A Neural Network Approach to Modeling Inverse Problems.”**Organizational Experience**

Organized mini-symposium at SIAM Texas-Louisiana Sectional Meeting

November 2023

Mathematical and Computational Foundations of Predictive Digital Twins

Co-organized with L. Cao, D. Luo

Awards and Honors

CNS Catalyst Grant 2023-2024
CSEM Graduate Student Fellowship 2018-2022
Turing Honors Scholar 2014-2018
Dean's Honors Scholar 2014-2018
Recipient, College of Natural Sciences Scholarship 2014
Recipient, Siemens Foundation Scholarship 2014
University Honors 2014-2018
College of Natural Sciences Honors 2015-2018
Regional Finalist at Siemens Competition in Math, Science, and Technology 2013

Professional Societies and Campus Organizations

Society for Industrial and Applied Mathematics (SIAM)
American Physical Society (APS)
Graduate Student Assembly – *Primary Representative*: Fall 2022 – Fall 2024
SIAM Applied Math Mentorship Program: Spring 2021, Fall 2021, Spring 2022
STEM MUSE Mentorship Program: Fall 2023, Spring 2024